*Universidade Federal de Santa Catarina*
*UFSC - Campus Blumenau*

# INTRODUÇÃO À ROBÓTICA INDUSTRIAL

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# MOTIVAÇÃO

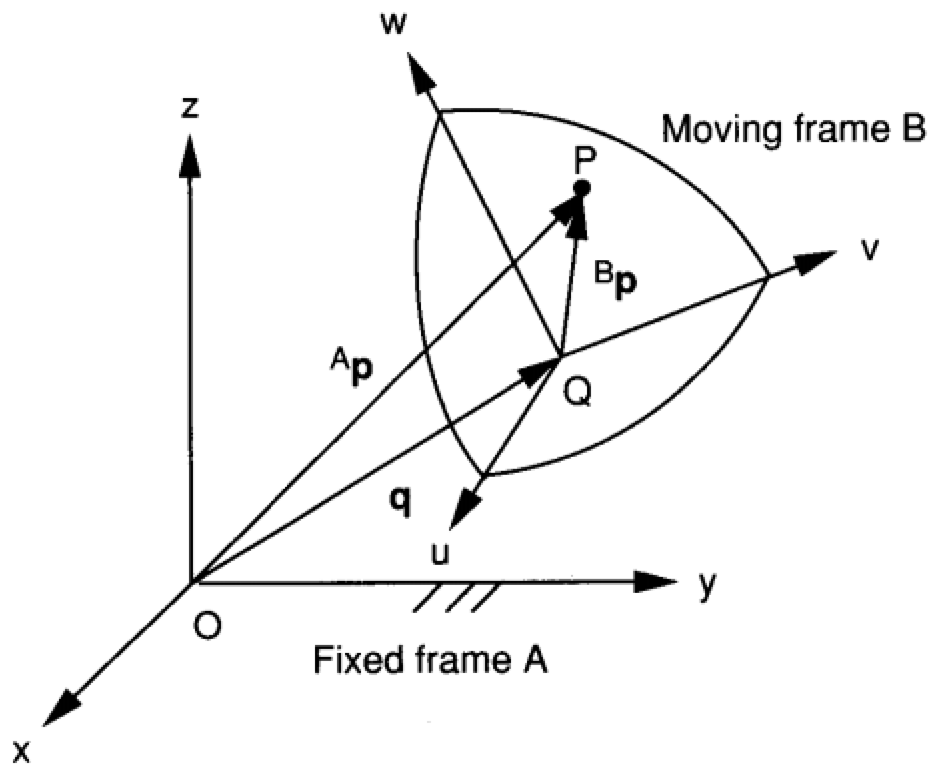
# Comportamento cinemático de sistemas mais complexos:

Alguns sistemas mecânicos como os robôs antropomórficos são mais complexos e a sua representação vetorial é difícil de ser obtida, nesses casos é necessário usar uma outra abordagem para modelar o comportamento desses sistemas:

# POSIÇÃO, ORIENTAÇÃO E LOCALIZAÇÃO DE UM CORPO RÍGIDO NO ESPAÇO

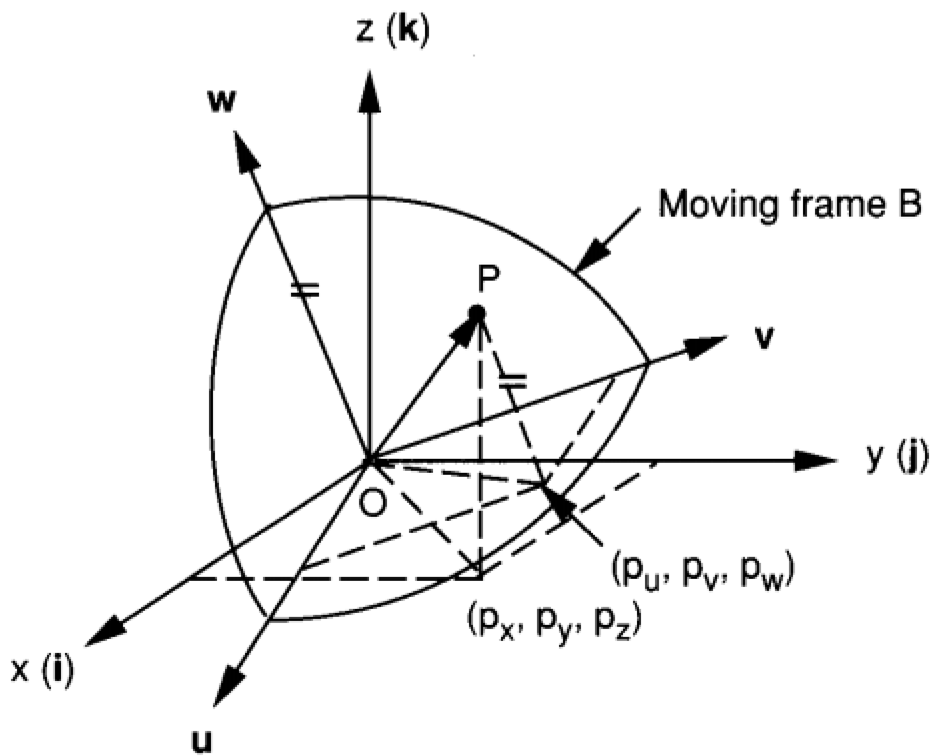
# Posição

A posição de um ponto no espaço em relação a um sistema de referência pode ser determinada usando um vetor de posição 3x1.

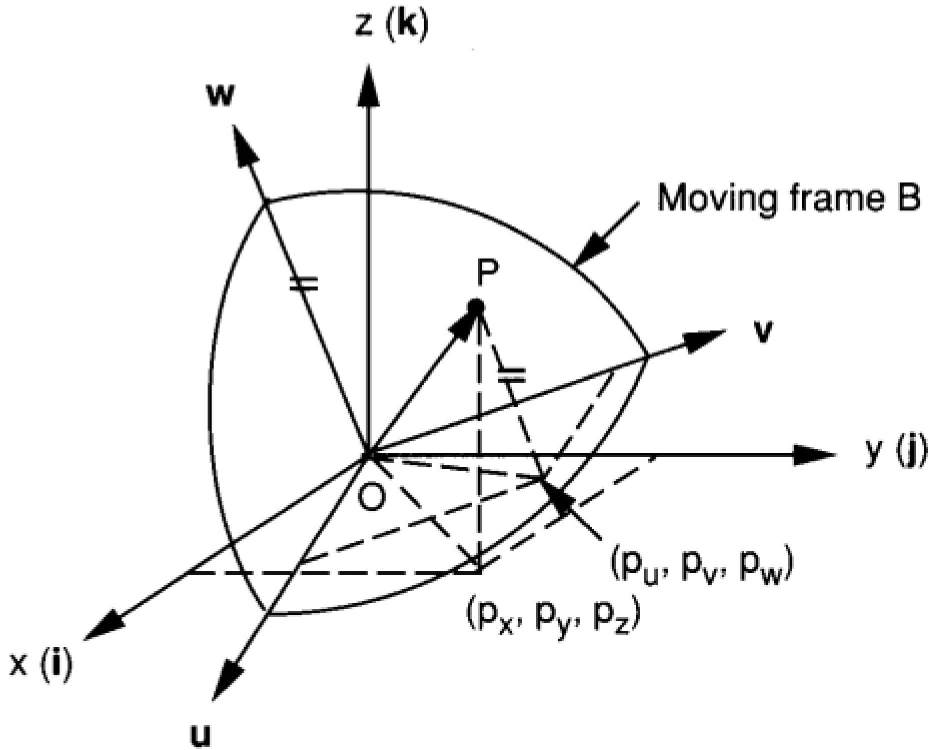


# Orientação

A orientação de um corpo rígido no espaço em relação a um sistema de referência pode ser descrita usando várias abordagens dentre as que se destacam: Os cosenos diretores, Os Angulos de Euler e Os Quaternions.



# Orientação usando cosenos diretores (Obtenção das matrizes de rotação):




$$^{A}\mathbf{p} = {}^{A}R_{B}\,{}^{B}\mathbf{p},$$

$$^{A}R_{B} = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

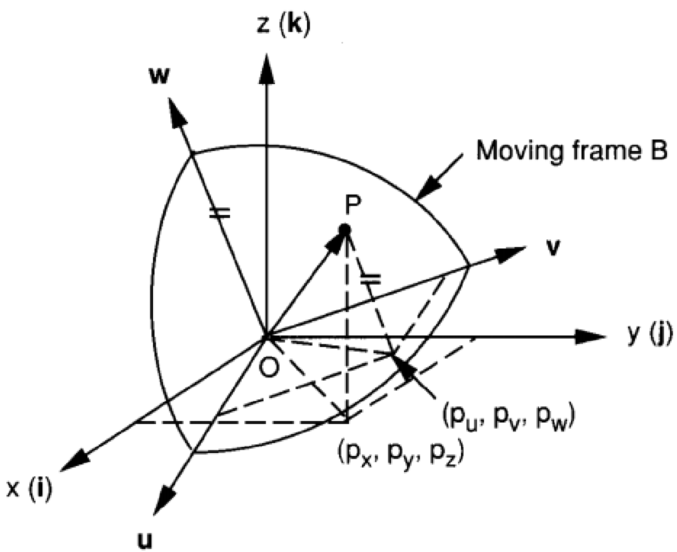
$$R(x,\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\psi & -s\psi \\ 0 & s\psi & c\psi \end{bmatrix}$$

$$R(y,\phi) = \begin{bmatrix} c\phi & 0 & s\phi \\ 0 & 1 & 0 \\ -s\phi & 0 & c\phi \end{bmatrix}$$

$$R(z,\theta) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

# PROPRIEDADES DAS MATRIZES DE ROTAÇÃO

**Propriedades das Matrizes de rotação:** Matrizes de rotação são constituídas por três vetores unitários em um sistema ortogonal, portanto as seguintes propriedades podem ser deduzidas.




$$^{A}R_{B} = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix}$$

$$\boxed{\begin{aligned} \mathbf{u}^2 &= 1, \\ \mathbf{v}^2 &= 1, \\ \mathbf{w}^2 &= 1, \end{aligned}}$$

$$\boxed{\begin{aligned} \mathbf{u}^{\mathrm{T}}\mathbf{v} &= 0, \\ \mathbf{v}^{\mathrm{T}}\mathbf{w} &= 0, \\ \mathbf{w}^{\mathrm{T}}\mathbf{u} &= 0. \end{aligned}}$$

$$\boxed{\begin{aligned} \mathbf{u} \times \mathbf{v} &= \mathbf{w}, \\ \mathbf{v} \times \mathbf{w} &= \mathbf{u}, \\ \mathbf{w} \times \mathbf{u} &= \mathbf{v}. \end{aligned}}$$

$$\boxed{\det(^{A}R_{B}) = 1}$$

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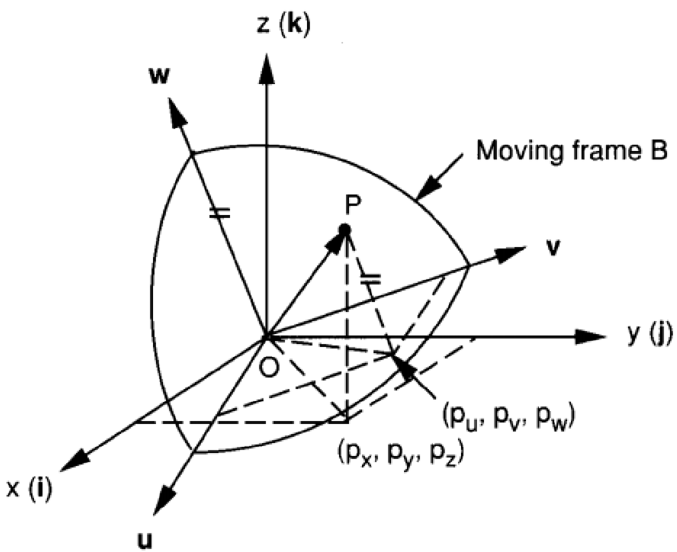


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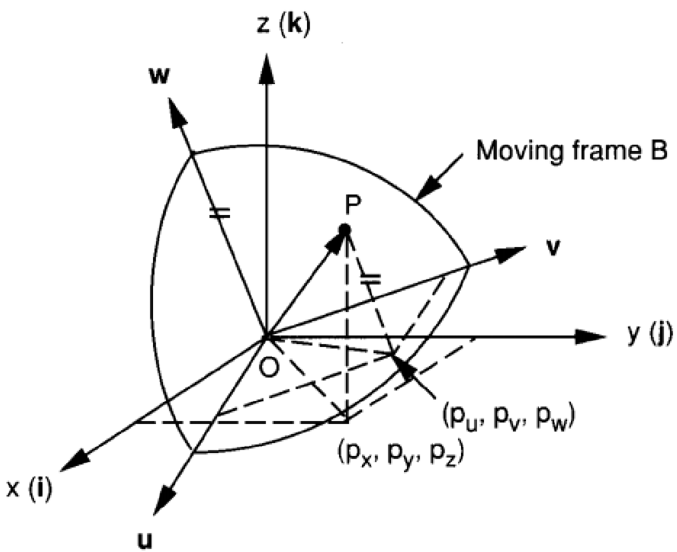


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**Exemplo:** Encontre todos os parâmetros da matriz de rotação mostrada abaixo.

$$R = \begin{bmatrix} ? & 0 & ? \\ 0.5 & ? & ? \\ 0 & ? & ? \end{bmatrix}$$

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**Exemplo:** rotacione múltiplos pontos de um corpo simultâneamente.

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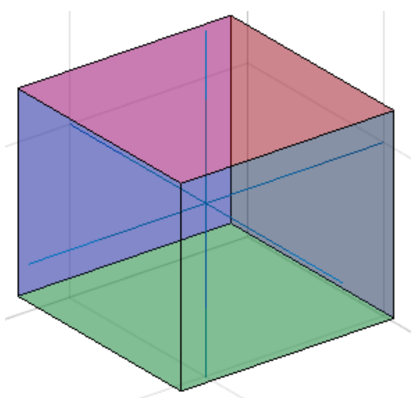


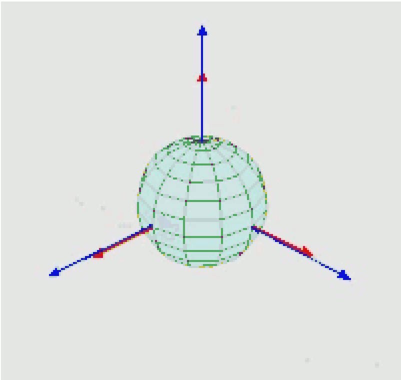
# ANGULOS DE EULER

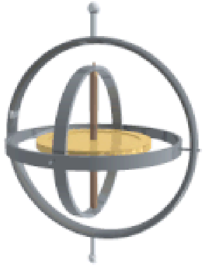
**Ângulos de Euler:** Em uma matriz de rotação usando cosenos diretores, 9 parâmetros são necessários. Contudo, um corpo rotacionando no espaço precisa apenas de 3 parâmetros para definir o seu comportamento.

$$^{A}\mathbf{p} = {}^{A}R_{B}\, {}^{B}\mathbf{p},$$

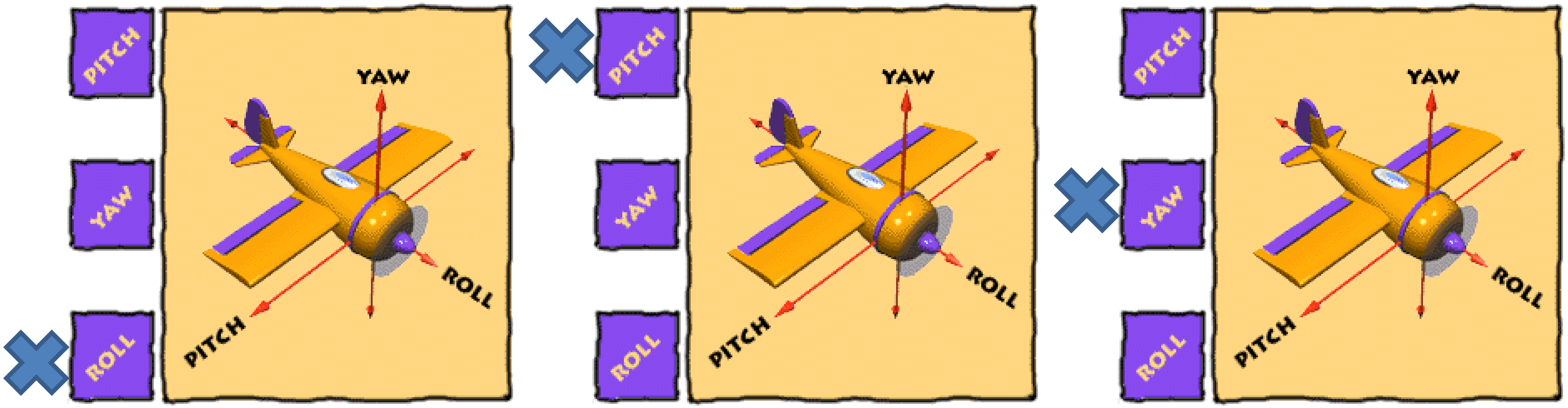
$$^{A}R_{B} = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Usando a representação usando os Ângulos de Euler, três rotações sucessivas sobre os eixos principais são necessárias.





## “Roll-Pitch-Yaw Angles” (rolamento, inclinação e guinada):




$$R(\psi, \theta, \phi) = R(z, \phi)\, R(y, \theta)\, R(x, \psi)$$

$$R(\psi, \theta, \phi) = \begin{bmatrix} c\phi c\theta & c\phi s\theta s\psi - s\phi c\psi & c\phi s\theta c\psi + s\phi s\psi \\ s\phi c\theta & s\phi s\theta s\psi + c\phi c\psi & s\phi s\theta c\psi - c\phi s\psi \\ -s\theta & c\theta s\psi & c\theta c\psi \end{bmatrix}$$

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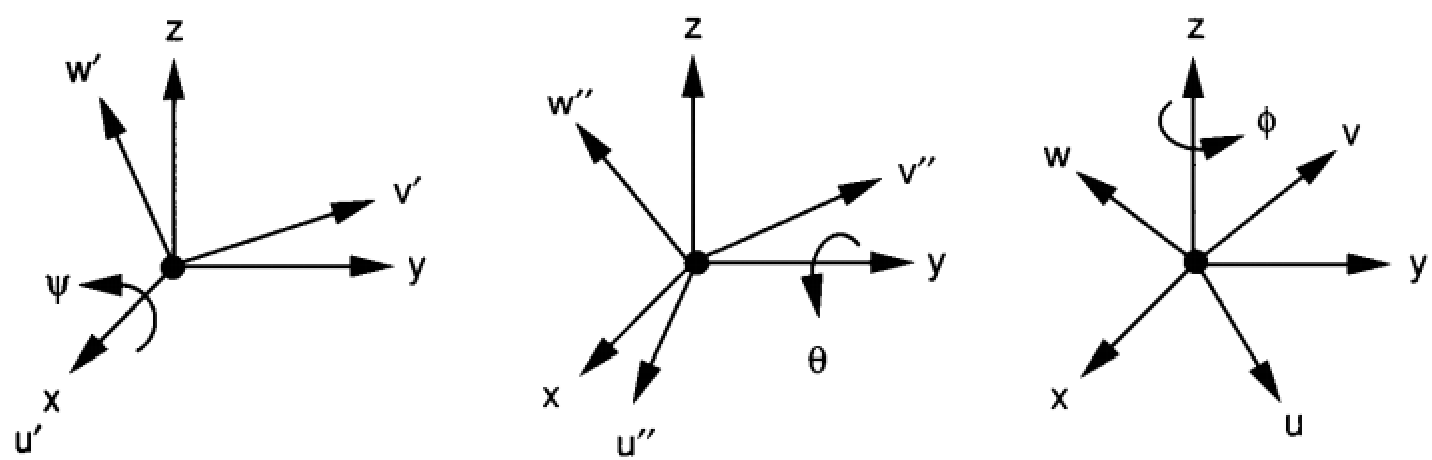

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$$R(\psi, \theta, \phi) = R(z, \phi)\, R(y, \theta)\, R(x, \psi)$$

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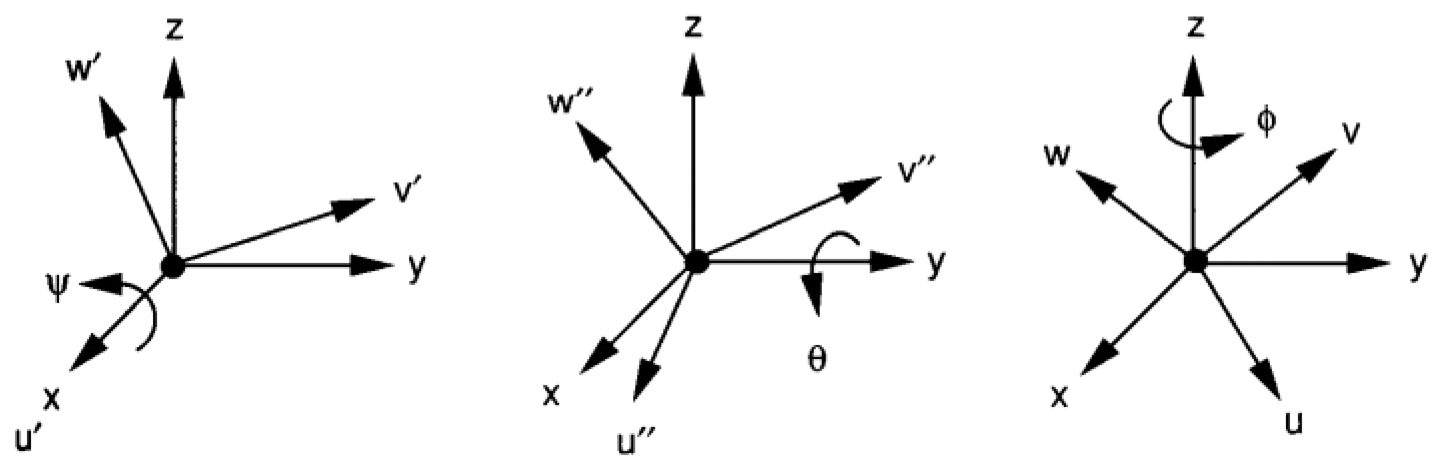
$$\theta = \sin^{-1}(-a_{31}),$$

$$\psi = \mathrm{Atan2}(a_{32}/\mathrm{c}\theta,\, a_{33}/\mathrm{c}\theta),$$

$$\phi = \mathrm{Atan2}(a_{21}/\mathrm{c}\theta,\, a_{11}/\mathrm{c}\theta),$$

# OUTRAS CONVENÇÕES PARA OS ANGULOS DE EULER



Existem outras convenções para obter matrizes com 3 rotações sucessivas (Sobre o sistema Fixo):

$$R_{X'Y'Z'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\beta c\gamma & -c\beta s\gamma & s\beta \\ s\alpha s\beta c\gamma + c\alpha s\gamma & -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta \\ -c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha s\beta s\gamma + s\alpha c\gamma & c\alpha c\beta \end{bmatrix},$$

$$R_{X'Z'Y'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\beta c\gamma & -s\beta & c\beta s\gamma \\ c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma \\ s\alpha s\beta c\gamma - c\alpha s\gamma & s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{Y'X'Z'}(\alpha,\beta,\gamma)=\begin{bmatrix} s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma & s\alpha c\beta \\ c\beta s\gamma & c\beta c\gamma & -s\beta \\ c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha c\beta \end{bmatrix},$$

$$R_{Y'Z'X'}(\alpha,\beta,\gamma)=\begin{bmatrix} c\alpha c\beta & -c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha s\beta s\gamma + s\alpha c\gamma \\ s\beta & c\beta c\gamma & -c\beta s\gamma \\ -s\alpha c\beta & s\alpha s\beta c\gamma + c\alpha s\gamma & -s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

$$R_{Z'X'Y'}(\alpha,\beta,\gamma)=\begin{bmatrix} -s\alpha s\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta & s\alpha s\beta c\gamma + c\alpha s\gamma \\ c\alpha s\beta s\gamma + s\alpha c\gamma & c\alpha c\beta & -c\alpha s\beta c\gamma + s\alpha s\gamma \\ -c\beta s\gamma & s\beta & c\beta c\gamma \end{bmatrix},$$

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$$R_{Z'X'Z'}(\alpha,\beta,\gamma)=\begin{bmatrix} -s\alpha c\beta s\gamma + c\alpha c\gamma & -s\alpha c\beta c\gamma - c\alpha s\gamma & s\alpha s\beta \\ c\alpha c\beta s\gamma + s\alpha c\gamma & c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha s\beta \\ s\beta s\gamma & s\beta c\gamma & c\beta \end{bmatrix},$$

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Existem outras convenções para obter matrizes com 3 rotações sucessivas (Sobre o sistema Móvel):

$$R_{XYZ}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix},$$

$$R_{XZY}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & -c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha s\beta s\gamma + s\alpha c\gamma \\ s\beta & c\beta c\gamma & -c\beta s\gamma \\ -s\alpha c\beta & s\alpha s\beta c\gamma + c\alpha s\gamma & -s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

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$$R_{YZX}(\gamma,\beta,\alpha) = \begin{bmatrix} c\beta c\gamma & -s\beta & c\beta s\gamma \\ c\alpha s\beta c\gamma + s\alpha s\gamma & c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma \\ s\alpha s\beta c\gamma - c\alpha s\gamma & s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma \end{bmatrix},$$

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**Exemplo:** rotacione múltiplos pontos de um corpo simultaneamente usando os Ângulos de Euler.

$$R(x,\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\psi & -s\psi \\ 0 & s\psi & c\psi \end{bmatrix}$$

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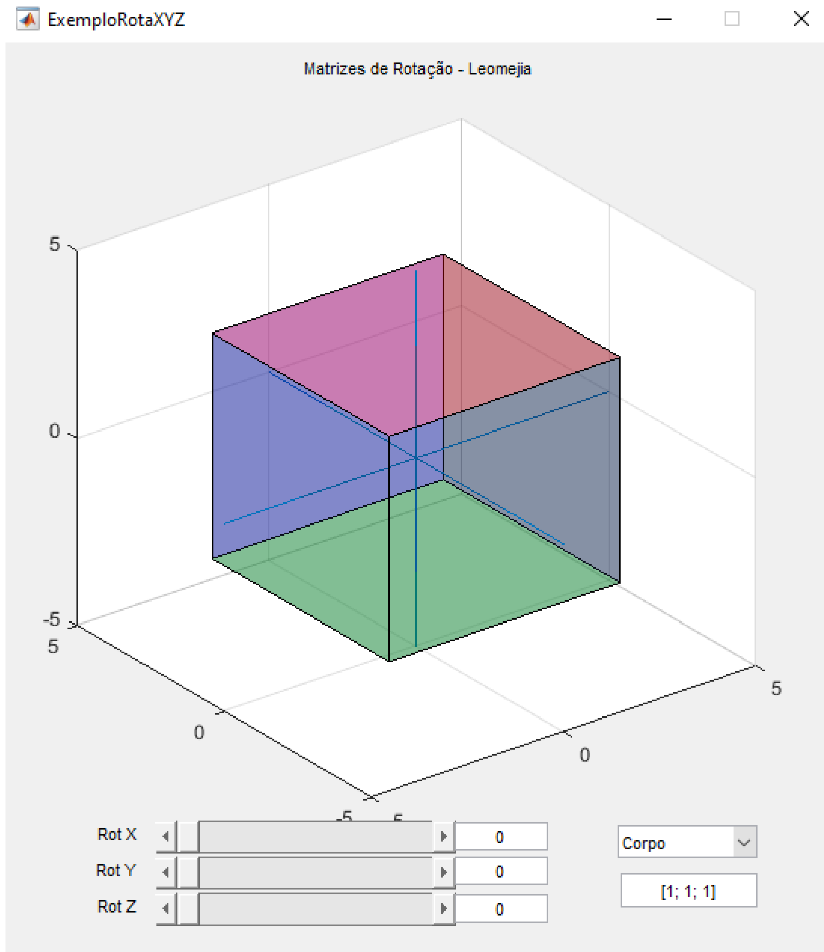
$$R(z,\theta) = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



# *Fim da Aula*

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